



## CS102-Computer Programming 1

### Programming Exercise 4

#### 1<sup>st</sup> semester 2025-2026

#### Objective:

- To develop students' problem-solving and algorithmic thinking skills in programming.
- To apply selection constructs (**if-else**, **switch**) in C and other fundamental programming concepts in the implementation of the algorithms.

For each of the problem, design the algorithm in flowchart and implement them in C language. Text in the boxes shows the sample output during the execution of the program.

#### Problem 1. Triangle Type Identifier

Create a C program that determines the type of triangle based on the lengths of its three sides. The program should classify the triangle as:

- Equilateral: if all sides are equal.
- Isosceles: if two sides are equal.
- Scalene: if no sides are equal.
- Not a Triangle: if the sum of any two sides is less than or equal to the third side (invalid triangle).

Sample outputs during execution:

```
Enter side 1: 5
Enter side 2: 5
Enter side 3: 5

Triangle type: Equilateral
```

```
Enter side 1: 3
Enter side 2: 4
Enter side 3: 10

Not a valid triangle.
```

#### Problem 2. Electricity Bill Calculator

Create a C program that calculates a user's electricity bill based on their power consumption (in kWh). The rate structure is:

- Up to 100 kWh: \$0.50 per kWh
- 101 to 300 kWh: \$0.75 per kWh
- Above 300 kWh: \$1.00 per kWh

Sample outputs during execution:

```
Enter name: John
Enter power consumption (kWh): 250

John's bill at $0.75/kWh is: $187.50
```

```
Enter name: Sarah
Enter power consumption (kWh): 350

Sarah's bill at $1.00/kWh is: $350.00
```

### Problem 3: Gym Membership Discount

Write a program that calculates the membership price of a gym based on age and type of membership using nested if-else.

Membership types:

- 1 → "Basic" membership: \$100
- 2 → "Premium" membership: \$500
- 3 → "VIP" membership: \$1000

Price discounts:

- If age is below 18, apply a 20% discount for any membership.
- If age is between 18 and 60, no discount.
- If age is above 60, apply a 30% discount for any membership.

Sample output during execution:

```
Enter age: 65
Enter membership type (1-3): 2

Price after discount: $350 (30% discount applied to Premium membership)
```

### Problem 4. BMI (Body Mass Index) Calculator with Health Categories

Write a C program that calculates the Body Mass Index (BMI) based on a person's height (in meters) and weight (in kilograms). The formula for BMI is:  $BMI = \frac{weight}{height^2}$

Based on the BMI value, the program should classify the person into one of the following categories:

- Underweight: BMI < 18.5
- Normal weight: BMI 18.5 - 24.9
- Overweight: BMI 25 - 29.9
- Obese: BMI ≥ 30

Sample output during execution:

```
Enter name: Alice
Enter weight (kg): 65
Enter height (m): 1.7
Alice's BMI: 22.49

Category: Normal weight
```

```
Enter name: Bob
Enter weight (kg): 100
Enter height (m): 1.75
Bob's BMI: 32.65

Category: Obese
```

### Problem 5: Employee Bonus Calculator

Write a program that calculates an employee's bonus based on their years of service and job position using selection.

Bonus structure:

- If years of service is less than 5, no bonus.
- If years of service is 5 or more:
  - If position is 1 (Manager) → 20% of salary.
  - If position is 2 (Team Lead) → 15% of salary.
  - If position is 3 (Staff) → 10% of salary.

Sample output during execution:

```
Enter years of service: 7
Enter position (1-3): 2
Enter salary: 50000

Bonus: 7500 (15% of salary for Team Lead)
```

### Note to Students:

- Make sure to apply the concepts discussed in the lectures, especially the selection **if-else** and **switch** statements.
- Be prepared to share and explain to the class your output on this exercise.
- Submit the following files:
  - **LASTNAME-FIRSTNAME\_PE4\_Algorithms.pdf** (contains the flowchart of the algorithms for all the problems)
  - **LASTNAME-FIRSTNAME\_PE4\_problem1.c** (source code for Problem 1)
  - **LASTNAME-FIRSTNAME\_PE4\_problem2.c** (source code for Problem 2)
  - **LASTNAME-FIRSTNAME\_PE4\_problem3.c** (source code for Problem 3)
  - **LASTNAME-FIRSTNAME\_PE4\_problem4.c** (source code for Problem 4)
  - **LASTNAME-FIRSTNAME\_PE4\_problem5.c** (source code for Problem 5)